PROJECT PLAN

Project:	Factory Inventory & Production Management System (MVP)
Release:	Release 1.0 (MVP)
Date:	15 February 2026
PM Approach:	Agile (Scrum) – 4 sprints
Author:	Uditha Balalla
Owner:	Uditha Balalla
Client:	Module Supervisor / Stakeholder
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Date of Next revision:

Revision date	Previous revision date	Summary of Changes	Changes marked
15 February 2026	—	Updated project plan with sprint roadmap and full role structure.	No changes were marked since this is the 1st revision

1.3 Approvals

This document requires the following approvals.
Signed approval forms are filed in the Management section of the project files.

Name	Signature	Title	Date of Issue	Version
Uditha Balalla		Project manager	15 February 2026	1.1

1.4 Distribution

This document has been distributed to:

Name	Title	Date of Issue	Version
Uditha Balalla	Project manager	15 February ...	1.1
Nisal Wickramaarachchi	Start-up manager	15 February ...	1.1
Dulanjali Kulathunga	Scheduling manager	15 February ...	1.1
H J M S M jayasinghe	Quality manager	15 February ...	1.1
Heshan Shalinda Koralagamage	Risk manager	15 February ...	1.1

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3 Purpose of Document

The purpose of this document is to define the project, to form the basis for its management and the assessment of overall success.

4 Background

A single food-production factory requires consistent control over raw material receiving, production batching, finished goods availability, and wholesaler dispatching. When these activities are handled through spreadsheets or manual notes, typical issues include stock mismatches, limited traceability for audits, expiry-related waste, and delays in generating dispatch documentation. This project introduces a centralised web system with role-based access, a stock ledger with traceability, FEFO dispatch allocation, reporting, and audit logs to improve operational accuracy and decision-making.

System Details: Monorepo web application for a single-factory food production workflow. Client: React SPA (Vite + TypeScript + Ant Design). Server: Express + TypeScript (Mongoose) deployed as a Vercel serverless function with handler api/[...path].ts. Authentication: Firebase Email/Password on client with Firebase Admin token verification on server. Database: MongoDB Atlas (replica set recommended for transactions). Core modules: users (OWNER-only), products, suppliers, wholesalers, GRN (draft/confirm stock-in), BOM, batches (DRAFT → IN_PROGRESS → COMPLETED), orders, dispatch (FEFO allocation + delivery note PDF), reports (stock, movements, expiry, production, wastage, sales, traceability), and audit (OWNER-only).

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5 Project Definition

5.1 Project Objectives

Project objectives (measurable outcomes):

- Deliver an MVP increment within 4 timeboxed sprints (planning, review, retrospective for each sprint).
- Maintain sprint schedule and meeting minutes in a shared drive (updated each sprint).
- Implement secure sign-in and enforce role-based access control (server-side) for OWNER/EMPLOYEE/WHOLESALER.
- Provide end-to-end operational workflows: GRN receiving/confirm, BOM management, production batches, orders, and dispatch.
- Implement FEFO-based allocation for dispatch and generate delivery note PDFs.
- Provide reports and audit trail for traceability and accountability (audit view restricted to OWNER).
- Maintain risk register and quality log; ensure Definition of Done is applied to sprint deliverables.

Planned work by sprint (brief):

- **Sprint 1 – Kickoff + foundations:** Repo/env setup, auth + RBAC baseline, base master data + inventory CRUD baseline, quality & risk artefacts, test framework baseline.
- **Sprint 2 – Stock + production foundations:** GRN confirm + stock movements, BOM management, batch lifecycle core (DRAFT → IN_PROGRESS), FEFO logic foundation + tests, DoD checklist adoption.
- **Sprint 3 – Orders + dispatch + traceability:** Orders flow, dispatch consumes FEFO stock, delivery note PDF, improved audit trail, reporting baseline, UAT/demo script.
- **Sprint 4 – Hardening + deployment + polish:** Validation and security hardening, deployment readiness, documentation finalisation, UI polish, final demo assets.

5.2 Defined Method of Approach

The project will be managed using Agile Scrum across 4 sprints. Work is tracked in a Product Backlog prioritised using MoSCoW (Must/Should/Could/Won't) and estimated using story points. Each sprint includes Sprint Planning (scope + estimates), daily coordination, backlog refinement, a Sprint Review (demo of the increment), and a Sprint Retrospective (process improvements). Quality gates (Definition of Done, code review, lint/tests) and a weekly risk review are applied throughout.

5.3 Project Scope

In scope:

- React SPA client and Express/TypeScript API deployed as two Vercel projects.
- Firebase Auth (Email/Password) with backend token verification via Firebase Admin.
- MongoDB Atlas persistence for master data and operational records.
- RBAC enforcement for OWNER, EMPLOYEE, WHOLESALER roles.
- Modules: users, products, suppliers, wholesalers, GRN, BOM, batches, orders, dispatch, reports, audit.
- Stock ledger and movement tracking to support traceability.
- FEFO dispatch allocation and delivery-note PDF generation.
- Documentation pack (setup, architecture, deployment, user guide) and sprint artefacts (risk/quality logs).

Out of scope:

- Multi-factory support, multi-warehouse optimisation, and advanced supply chain planning.
- Accounting/ERP integrations, invoicing, and payment handling.
- Offline-first mobile application support.
- Advanced forecasting/AI procurement recommendations beyond reporting.

5.4 Project Deliverables and/or Desired Outcomes

Overall deliverables:

- Sprint plans (4 sprints) with assigned owners and estimates.
- Meeting minutes and schedule updates maintained by Scheduling Manager.
- Working MVP system increment delivered at the end of each sprint (demo-ready).
- Authentication + RBAC enforcement.
- Operational modules (GRN, BOM, batches, orders, dispatch) with stock ledger and audit logs.
- Reports module and delivery-note PDF generation.
- Documentation pack: README, Setup guide, Architecture, Deployment guide, User guide.
- Risk Management Plan + Risk Register; Quality Plan + Quality Log.

Sprint	Sprint goal	Key outcomes (planned)
Sprint 1	Kickoff + foundations	Repo/env setup, auth + RBAC baseline, base master data + inventory CRUD baseline, quality & risk artefacts, test framework baseline.
Sprint 2	Stock + production foundations	GRN confirm + stock movements, BOM management, batch lifecycle core (DRAFT → IN_PROGRESS), FEFO logic foundation + tests, DoD checklist adoption.
Sprint 3	Orders + dispatch + traceability	Orders flow, dispatch consumes FEFO stock, delivery note PDF, improved audit trail, reporting baseline, UAT/demo script.
Sprint 4	Hardening + deployment + polish	Validation and security hardening, deployment readiness, documentation finalisation, UI polish, final demo assets.

5.5 Exclusions

Exclusions (not included):

- Multi-factory support, multi-warehouse optimisation, and advanced supply chain planning.
- Accounting/ERP integrations, invoicing, and payment handling.
- Offline-first mobile application support.
- Advanced forecasting/AI procurement recommendations beyond reporting.

5.6 Constraints

Constraints:

- Timeboxed to 4 sprints with a small project team (named roles above).
- Fixed technology stack: React/Vite client, Express/TypeScript server, Firebase Auth, MongoDB Atlas, Vercel deployment.
- MongoDB transaction support depends on using a replica set configuration.
- CORS and environment configuration must be correct for deployed environments.

5.7 Assumptions

Assumptions:

- A Firebase project with Email/Password authentication is available for development and demo accounts.
- A MongoDB Atlas cluster is available (replica set recommended).
- Vercel deployments for client and server are permitted for demo/review.
- Stakeholder/supervisor feedback is available at least once per sprint review.
- All team members can access the shared drive for artefacts and minutes.

5.8 Legal and ethical

Legal and ethical considerations:

- Role-based access control and least privilege must protect operational and user data.
- Secrets (Firebase service account, MongoDB URI) must be stored in environment variables and never committed to version control.
- If personal data is stored (users/contacts), it must be handled responsibly in line with data protection principles.
- Third-party libraries must be used under their license terms and updated where practical.
- Team members follow professional conduct (respectful communication, integrity, accountability).

6 Project Organisation Structure

6.1 Project Management Team Structure

Project management team structure (roles and responsibilities):

Role	Name	Key responsibilities	Reports / updates to
Project manager	Uditha Balalla	Overall project direction, approvals, escalation decisions, stakeholder reporting.	Stakeholders
Start-up manager	Nisal Wickramaarachchi	Backlog ownership, sprint scope, integration coordination, demo readiness, documentation oversight.	Project mana...
Scheduling manager	Dulanjali Kulathunga	Sprint calendar, meeting scheduling, minutes, schedule updates and reminders.	Project mana...
Quality manager	H J M S M jayasinghe	Quality planning, DoD, review checklist, test strategy, quality log, QA sign-off.	Project mana...
Risk manager	Heshan Shalinda Koralagamage	Risk plan, risk register, mitigation tracking, security/reliability checks.	Project mana...

6.2 Individual Role Descriptions

Project manager – Uditha Balalla

- Owns the Project Plan and approvals (versioning, distribution, and change control).
- Monitors sprint progress and resolves escalations and resource conflicts.
- Ensures alignment with module expectations and coordinates stakeholder communications.

Start-up manager – Nisal Wickramaarachchi

- Maintains the Product Backlog, priorities (MoSCoW), and sprint scope proposals.
- Coordinates integration of modules and ensures sprint demo readiness.
- Maintains key documentation artefacts and ensures deliverables are stored in Drive.

Scheduling manager – Dulanjali Kulathunga

- Maintains the sprint schedule/calendar and meeting invitations.
- Records and shares meeting minutes (planning, review, retrospective).
- Tracks action items and follow-ups with owners and due dates.

Quality manager – H J M S M jayasinghe

- Defines Definition of Done and sprint quality gates.
- Ensures reviews, linting, and tests are completed before merge/acceptance.
- Maintains the Quality Log and provides QA sign-off for sprint review demos.

Risk manager – Heshan Shalinda Koralagamage

- Defines risk scoring/categories and runs weekly risk review cadence.
- Maintains the Risk Register (owners, mitigations, status) and triggers actions.
- Performs security/reliability checks (RBAC, secrets handling, deployment configuration).

7 Communication Plan

Communication plan (summary):

Communication	Purpose	Participants	Frequency	Owner
Daily stand-up (short)	Blockers, progress, next steps	All roles	Daily (wo...	Nisal Wic...
Sprint planning	Confirm sprint goal, scope, estimates	PM + Sta...	Start of e...	Uditha B...
Backlog refinement	Re-prioritise and clarify upcoming items	PM + Sta...	Mid-sprint	Nisal Wic...
Quality checkpoint	DoD adherence, defects, test status	Quality + ...	Mid-sprint	H J M S ...
Risk review	Update top risks, mitigations, actions	Risk + P...	Weekly	Heshan ...
Sprint review (demo)	Show increment and collect feedback	All roles ...	End of ea...	Uditha B...
Sprint retrospective	Process improvements and action items	All roles	End of ea...	Dulanjali ...

8 Project Quality Plan

Project quality plan (summary):

- Definition of Done applied to all sprint deliverables (acceptance criteria, review, lint/tests, docs, demo-ready).
- Code review required before merge; automated tests must pass for accepted items.
- Sprint Quality Log maintained by the Quality Manager with severity and actions.
- Security/reliability checks performed with the Risk Manager (RBAC, secrets, deployment/CORS).

Quality artefacts are stored in the project drive under the Quality folder (Quality Plan + Quality Log).

9 Project Controls

Project controls (direction, monitoring, and reporting):

- **Schedule control:** Sprint calendar maintained; meetings scheduled; minutes recorded; action items tracked. Owner: Dulanjali Kulathunga.
- **Progress control:** Backlog and sprint board updated with story points; sprint review demonstrates increment. Owner: Uditha Balalla.
- **Change control:** Change requests logged; PM approves scope changes; de-scope to maintain sprint goal if needed. Owner: Uditha Balalla.
- **Quality control:** DoD checklist, PR reviews, lint/tests required; QA sign-off before sprint review. Owner: H J M S M jayasinghe.
- **Risk control:** Risk register updated weekly; top risks reviewed and mitigations assigned with owners. Owner: Heshan Shalinda Koralagamage.
- **Release control:** Demo-ready increment required; deployment checklist followed; smoke flow executed. Owner: Nisal Wickramaarachchi.

Exception process: If a sprint goal is at risk, the Project Manager calls an exception discussion to re-scope or re-plan the sprint.

10 Initial Risk Log

Initial risk log (top risks):

Risk ID	Risk	Category	Probability (1-5)	Impact (1-5)	Score	Owner	Mitigation (summary)
R-01	CORS allowlist misconfigured blocks browser API calls	De...	3	4	12	He...	Use exact origin allowlist + preflight handling; verify with curl; redeploy.
R-02	Vercel route mapping misses nested /api paths causing 404s	De...	2	4	8	He...	Ensure /api/(.*) routes to api/[...path].ts; validate endpoints post-deploy.
R-03	Firebase private key formatting causes auth failures	Te...	3	4	12	Ni...	Ensure \n newlines preserved; add startup health check for Firebase Admin

							init.
R-04	RBAC bypass exposes restricted endpoints	Se…	2	5	10	He…	Centralise require Role; add authz tests; route review before release.
R-05	FEFO allocation logic defect dispatches wrong batches	Qu…	3	5	15	H …	Unit tests + edge cases; peer review; fallback manual allocation if needed.
R-06	Negative stock due to missing validation / concurrency	Da…	3	4	12	H …	Server-side guards + atomic updates; tests for deductions.
R-07	Secrets committed to git repository	Se…	2	5	10	He…	.gitignore + checklist; rotate keys immediately if leaked.

R-08	Schedule/capacity constraints delay sprint goals	Sc...	3	4	12	Ud...	Capacity planning; timebox spikes; de-scope non-MVP items.